2.3.3 $(a \times b) \times c = a \times (b \times c)$. (Proposition 2.3.5)

Fixing a and c

Base cases $b = 0 \to (a \times b) \times c = 0 \times c = 0$;

$b = 0 \to a \times (b \times c) = a \times 0 = 0$,

$b = 1 \to (a \times b) \times c = (a) \times c$,

$b = 1 \to a \times (b \times c) = a \times (c)$

Inductively assuming we've proven $(a \times b) \times c \stackrel{!}{=} a \times (b \times c)$, for $b++$ we have

$(a \times b++) \times c = (b++ \times a) \times c = (b \times a + a) \times c = (a \times b \times c + a \times c)$,

$a \times (b++ \times c) = a \times (b \times c + c) = (a \times b \times c + a \times c)$